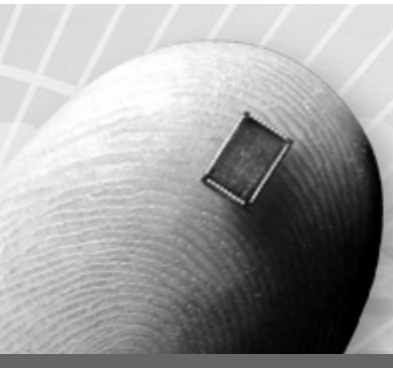
ON Semiconductor®



Evaluation and Development Board Manual

for BelaSigna® 300

M-20738-004

ON Semiconductor

March 2010

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1 Introduction

1.1 Purpose

The Evaluation and Development Board consists of two printed circuit boards: a motherboard and a Rapid Prototyping Module (RPM). They are designed for use with the software tools included with the Evaluation and Development Kit to evaluate the performance and capabilities of the BELASIGNA® 300 audio processing system. This manual describes all functional features and many use cases of the Evaluation and Development Board. It also describes how the boards work, and how they can be configured.

1.2 Intended Audience

This manual is intended for engineers and developers who will use the Evaluation and Development Board, with or without the motherboard component.

The manual assumes that the reader is familiar with BelaSigna 300 technology and tools, as well as using hardware evaluation boards. Anyone who wants to experience BelaSigna 300 audio processing algorithms, and turn a portable audio product into a DSP-enabled version, will benefit from using the Evaluation and Development Board and this manual.

1.3 Conventions

This manual displays the following names in a `monospace font`:

- Headers
- Connectors
- Potentiometers
- Switches

These items are labelled on the Evaluation and Development Board or shown on the schematics.

1.4 Manual Organization

The *Evaluation and Development Board Manual* contains the following chapters and appendices:

- **Chapter 1: Introduction**, describes the purpose of the manual, outlines how the manual is organized, and lists suggested documents for more information.
- **Chapter 2: Overview**, provides a high-level overview of the Evaluation and Development Board, and the features it provides in different use cases.
- **Chapter 3: Getting Started**, explains how to start using the hardware, provide a power supply, apply an audio input signal, and listen to the output.
- **Chapter 4: Rapid Prototyping Module**, provides detailed information on how to use the RPM as a standalone module, typically when no Evaluation and Development Board motherboard is available.

- **Chapter 5: Motherboard**, shows the benefits of using the RPM docked onto the motherboard for easier access to the main features of the RPM.
- **Appendix A: Configuration Options**, lists all the different accessories on the motherboard with their configuration options, like headers, switches, potentiometers and so on.
- **Appendix B: Schematics**, provides detailed schematics of both the RPM and the motherboard of the Evaluation and Development Board.
- **Appendix C: Layout**, provides a mechanical board representation of the RPM and the motherboard.

1.5 FURTHER READING

For more information, refer to the following documents:

- *Hardware Reference Manual for BelaSigna 300*
- *Firmware Reference Manual for Ezairo 5900 and BelaSigna 300*
- *Integrated Development Environment User's Guide*
- BelaSigna 300 data sheet

2 Overview

2.1 INTRODUCTION

The Evaluation and Development Board enables developers to experience the effect of BelaSigna 300 in the next generation of their portable audio devices.

The RPM can be integrated with existing products to experience the effect in place. It can also be accompanied by the motherboard, turning the Evaluation and Development Board into a portable BelaSigna 300 technology demonstrator. The motherboard provides easy access to all analog inputs and outputs, as well as most digital interfaces. It provides a convenient power supply and configuration options to the RPM.

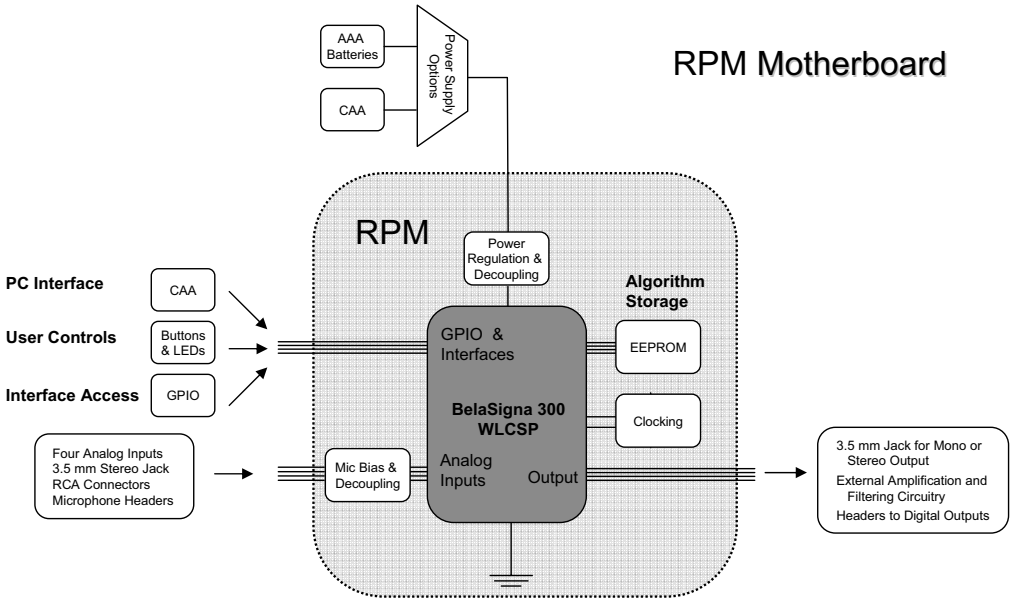


Figure 1: A Functional View of the Rapid Prototyping Module and Its Motherboard

2.2 RPM AT A GLANCE

The RPM provides advantages such as:

- Small form factor (20mm x 20mm)
- Non-volatile memory for algorithm storage
- On-board voltage regulation (any source from 1.7V to 5.5V)
- Fast 38.4MHz on-board clocking
- All required passive components mounted on board

2.3 MOTHERBOARD AT A GLANCE

The motherboard provides:

- Multiple power supply options
- Portable, battery-powered operation
- RCA audio inputs
- RCA audio output for testing the digital output using the on-board RC filter
- 3.5mm stereo jack connectors for audio inputs and outputs
- Easy communication with PC and ON Semiconductor software and evaluation tools using the CAA
- Access to most commonly-used digital interfaces
- Handy user interface options with buttons, potentiometers and LEDs
- External stereo digital-to-analog conversion and amplification
- Output signal filtering options

BelaSigna 250 is on the motherboard to function as a stereo PCM DAC only. This board is not designed as a BelaSigna 250 evaluation and development board.

3 Getting Started

A high sampling rate full-stereo application is preloaded in the RPM's EEPROM. In this default application, BelaSigna 300 samples the analog inputs `AI0` and `AI2` at a sampling frequency of 48kHz. The sampled data is transferred to the PCM interface and to the monaural direct drive output. An on-board DA converter and an audio amplifier enables the use of a standard headphone to hear the stereo output produced by BelaSigna 300. The user can define how the inputs are routed to the mono direct drive output (by default, the `AI0` input is routed to this output). The direct drive output allows the evaluation of the native sound quality available on BelaSigna 300.

3.1 With Motherboard

When the RPM is docked onto its companion motherboard, you can hear BelaSigna 300 processing your selected music. The following procedure configures the monaural direct drive output option, allowing you to verify that the setup is correct:

1. Provide power to the RPM motherboard using two AAA batteries. Other power supply options are available on the RPM motherboard; see Section 5.2, "Power Supply Options" on page 11 for more details.
2. To ensure that the whole board gets its power from the AAA batteries, use the configuration described in Table 1:

TABLE 1: CONFIGURING THE POWER

Header	Configuration
PSU-SEL	short pins 1 to 2
PSU-EN	short pins 1 to 2
VBAT-EN	open

3. Turn on the power on the board by moving switch `SW1` to the ON position, and check that LED `D1` lights up.
4. Connect a stereo analog signal from a music player or PC with a 3.5mm jack to the `MIC` connector.
5. Configure the output from the motherboard as described in Table 2:

TABLE 2: CONFIGURING THE OUTPUT

Header	Configuration
SHIELD-SEL	short pins 1 to 2
OUT-SEL	open pins 1 to 2
	open pins 3 to 4
	short pins 5 to 6
	short pins 7 to 8

TABLE 2: CONFIGURING THE OUTPUT (CONTINUED)

Header	Configuration
RCVR-SEL	open pins 1 to 2
	open pins 3 to 4
	short pins 5 to 6
	short pins 7 to 8
LC-FILT-EN	short pins 1 to 2
	short pins 3 to 4

Connect a headphone with a 3.5mm jack connector to the OUT connector to listen to the mono output of BelaSigna 300, provided on both the left and the right ear-cups of your headphone. Note that the Vol potentiometer has no effect when listening to the mono output signal; however, you can use the Rgain potentiometer to reduce the output level of the mono output. To listen to the externally amplified stereo signal, refer to Section 5.4.2, “Audio Output Stage” on page 14. This allows you to listen to the audio output of BelaSigna 250 via an audio amplifier located on the motherboard.

3.2 WITHOUT MOTHERBOARD

If you have an RPM without its companion motherboard, you must find a way to do the following before you can experience the sound quality of BelaSigna 300:

- Provide a power supply source
- Connect to a ground pin
- Connect input signals
- Connect speakers or headphones to the outputs

The above information is not within the scope of this manual. Contact your ON Semiconductor representative for assistance.

4 Rapid Prototyping Module

4.1 How to Integrate the RPM into Your Application

Most of the BelaSigna 300 pins are available on the RPM to interface with the host hardware. Four headers on the board provide connections. These headers are placed on the board in a way that also allows you to dock the RPM onto its companion motherboard.

Table 3 shows the four headers and the signals that are available on them.

TABLE 3: AVAILABLE SIGNALS ON THE FOUR HEADERS

Header X1: Available pins		Header X2: Available pins		Header X3: Available pins		Header X4: Available pins	
1	PCM_FR	1	I2C_SCL	1	AI0	1	RCVR_HP+
2	GPIO[0]/UART-TX	2	VBAT	2	PSU-IN	2	RCVR-
3	PCM_SERO	3	I2C_SDA	3	VREG	3	RCVR_HP-
4	GPIO[1]/LSAD[1]/UART-RX	4	VBAT	4	AI2	4	RCVR+
5	PCM_SERI	5	GNDD	5	AI1	5	RCVRGND
6	GPIO[2]/LSAD[2]	6	GNDD	6	PSU-IN	6	GND
7	GNDD	7	EXT_CLK	7	GNDA	7	RCVRGND
8	GPIO[3]/LSAD[3]	8	GNDD	8	AI4	8	GND
9	PCM_CLK			9	GNDA		
10	GPIO[4]/LSAD[4]			10	GNDA		

4.2 Power Supply Options

The default way to supply power to the RPM is to use on-board voltage regulation to provide a clean supply to BelaSigna 300. Alternatively, you can bypass the voltage regulator, and provide a direct supply to BelaSigna 300. This section discusses these two methods.

4.2.1 On-Board Voltage Regulation

The RPM has an on-board voltage regulator (Texas Instruments TPS73118 Low-Power Low-Dropout Regulator). This device uses the voltage provided on the PSU-IN pin, which must be in the range of 1.7V to 5.5V or V_{OUT} - V_{DO} where V_{DO} = 100mV maximum. PSU-IN is regulated down to 1.8V in a clean and optimal way to provide BelaSigna 300 with an adequate power supply on its VBAT pin. This is the default mode of operation; do not apply a power supply directly on the VBAT pin without applying hardware modifications, as described in Section 4.2.2, “Direct Power Supply” on page 8.

4.2.2 Direct Power Supply

If the host system can provide a clean and regulated 1.8V supply, you can disable the RPM's on-board regulator. To provide this externally regulated 1.8V supply directly on the VBAT pin, connect the PSU-IN pins (header X3, pins 2 and 6) to ground, which disables the voltage regulator. Refer to Figure 2 on page 28 for the detailed schematics.

4.2.3 Connecting the Ground Pins

Connect all ground pins of the RPM to the star ground point located at the output of the LDO. On the RPM, ensure the R2, R3 and R4 zero-ohm resistors (i.e., wires) are populated (they are, by default), or make external connections to the GNDA, GNDO and RCVRGND pins. This must be done in *all* cases of using the RPM without its companion motherboard.

4.3 CLOCKING OPTIONS

Typical applications of BelaSigna 300 require higher clock frequencies. For that reason, an on-board oscillator circuit provides a system clock frequency of 38.4MHz, providing significantly more computational power. From this system clock frequency, common sampling frequencies can be derived by configuring the on-chip clock dividers as shown in Table 4.

TABLE 4: CONFIGURING CLOCK DIVIDERS FOR VARIOUS SAMPLING FREQUENCIES

Sampling Frequency	ADC_CFG_ SAMPLE_FREQ	SLOWCLK	CLK_CFG1_SLOWCLK_ PRESCALE	System Clock & External Clock[1]
8 kHz	0x0C	1.28 MHz	30	38.4 MHz
16 kHz	0x02	1.28 MHz	30	38.4 MHz
32 kHz	0x02	2.56 MHz	15	38.4 MHz
48 kHz	0x02	3.84 MHz	10	38.4 MHz

1. Assuming that EXT_CLK is the same frequency as SYSCLK.

When running BelaSigna 300 at higher frequencies, the system current consumption increases. When the oscillator is running, the RPM power consumption also increases.

When you use a clock frequency other than the 38.4 MHz provided on the RPM board, you can remove the R5 resistor to disconnect the oscillation circuit from its power supply. Remove resistor R10 to isolate the EXT_CLK pin from this circuit, and provide a different clock source at this point.

IMPORTANT: BelaSigna 300 requires an external clock to boot, regardless of whether the on-board oscillator is disabled.

Refer to Figure 2 on page 28 for the on-board oscillator schematic.

4.4 BELASIGNA 300

A 3.63mm x 2.68mm 35-pin WLCSP package of BelaSigna 300 is used on the RPM. All the required passive elements like decoupling capacitors have also been added to the RPM design to make it easy to deploy.

4.5 EEPROM

A 512 kbit EEPROM is provided for algorithm storage. The EEPROM is directly powered by the VBAT pin of BelaSigna 300.

4.6 INTERFACES

4.6.1 Audio Input Stage

Connection points are provided on the RPM for each of the four audio inputs; they are available on header X3 (see Table 3 on page 7 for more details). These four connection points are all wired to BelaSigna 300 through a 100nF serial capacitor for DC filtering purposes. Typical headset microphones have two terminals. A DC bias must be present on the positive terminal to operate microphones requiring a bias voltage. The RPM design is configured to provide a microphone biasing circuit for inputs AI0 and AI2, whereas inputs AI1 and AI4 are simply provided on the connection points. Note that the chip input pin AI3 is connected to GNDA internally to the chip.

There are two options for microphone bias schemes on the RPM (AI0 and AI2):

- A 2.2 kΩ pull-up resistor to VREG (regulated 1V supplied by BelaSigna 300) to the microphone input. This is the default option and the resistors are present on the RPM.
- A voltage divider with an additional capacitor for filtering. Generally, the microphone has a lower sensitivity using the 2.2kΩ resistor from VREG due to the lower bias voltage, but you can often compensate for this with the adjustable preamp gains in BelaSigna 300 (0dB to 30dB). In cases where the sensitivity is too low, you can use the voltage divider network. Also, some microphones require higher bias voltage. The RPM was designed so that this voltage divider can be connected to either PSU-IN (3V) or VBAT (1.8V), depending on the requirements and types of microphone being used.

4.6.2 Audio Output Stage

The audio output has been brought out to connection points. The digital audio output (RCVR+/-) can drive most speakers directly (up to 25mW and down to 16Ω), with no external components. RCVR_HP+/- can handle audio outputs down to 8Ω. The four signals are available on header X4. See Table 3 on page 7 for details.

4.6.3 I²C Communication Interface

The I²C port provides full access to the chip for communicating with and debugging BelaSigna 300 devices. This port is supported by our development tools, which are part of the BelaSigna 300 Evaluation and Development Kit.

The I2C_SCL and I2C_SDA pins are wired to the connection header X2 for easy access on the RPM. Use standard communication interfaces available from ON Semiconductor, such as the Communication Accelerator Adaptor (CAA) to communicate with BelaSigna 300.

4.6.4 General-Purpose I/O

All 5 GPIO pins in BelaSigna 300 are brought out in header X1. Most of the GPIOs are multiplexed with other functions that can be enabled or disabled in software.

See Table 3 on page 7 for details on the headers.

4.6.5 PCM/I^{2}S Interface

The PCM/I^{2}S interface is often used for digital audio data communication; its four lines are provided on the X1 connection header as well.

5 Motherboard

5.1 RPM and Motherboard Interface

Using the RPM with its companion motherboard enables you to use the RPM without setting up more hardware. The motherboard is designed to provide access to the most commonly used signals on BelaSigna 300 through appropriate connectors on the motherboard. It can also act as a portable technology demonstrator to test signal processing algorithms running on the BelaSigna 300 audio processor.

The motherboard has four male headers that plug into corresponding female headers on the RPM. Once inserted, these connectors ensure connectivity between the two modules, and provide a stable way of attaching them together.

5.2 Power Supply Options

Various power supply methods were designed to support a variety of use models with the motherboard.

This design has two main power supply domains. The main motherboard requires 3V to operate, whereas the BelaSigna 300 device requires 1.8V to operate.

This section describes the different methods of generating the main 3V. The 3V power used on the motherboard is also provided to the RPM, which regulates it down to 1.8V for its own use. Provisions have also been made to directly power the RPM with 1.8V, thus disabling the RPM 1.8V regulator.

A main power switch controls the general power supply of the whole system (`SW1`), and the `PSU-SEL` header selects from the two different methods to power the board with 3V. The two default methods of getting power are the CAA supply where an on-board 3V charge pump is used, or two AAA batteries to provide the 3V. The `PSU-EN` header controls the actual supply voltage that the motherboard provides to the RPM. The LED `D1` indicates whether power has been correctly provided to the motherboard.

In most supply options described in this section, the `PSU-EN` header must short pins 1 and 2, providing a 3V power supply to the RPM, which regulates it down to 1.8V. The other configuration option on header `PSU-EN` (shorting pins 2 and 3) allows the motherboard to disable the RPM's voltage regulator. This option is discussed further in Section 5.2.2, "Communication Accelerator Adaptor (CAA) Power" on page 12.

The different options for supplying power to the microphones are described in Table 5.

TABLE 5: CONFIGURING MIC-PSU

Options	Configuration
VREG bias	short pins 1 to 2
1.8V bias	short pins 3 to 4
3V bias	short pins 5 to 6

5.2.1 AAA Batteries

Connecting pins 1 and 2 on the PSU-SEL header allows you to provide the main 3V power supply with two 1.5V AAA batteries (not included). This option allows the motherboard to be fully portable, enabling field testing of BelaSigna 300 algorithms.

5.2.2 Communication Accelerator Adaptor (CAA) Power

The ON Semiconductor CAA serves two purposes for the motherboard: it can provide power to the whole system, as well as providing the communication interface to BelaSigna 300. This section describes the power supply capability of the CAA, whereas the communication feature is described in Section 5.4.3, “I2C Communication Interface” on page 15.

To configure the CAA power supply, use the Integrated Development Environment (IDE) software tool, as described in the *Integrated Development Environment User’s Guide*.

Note: Do not continue to power the evaluation and development board from the CAA when doing audio quality measurements or qualification.

5.2.2.1 Use the CAA to Provide Power to the Motherboard

Connect pins 3 and 4 on the PSU-SEL header to provide the 3V main power supply for the motherboard by an on-board charge pump. The ON Semiconductor CAA is capable of providing a 1.8V supply through its 6-pin mini DIN cable on pin CAA-PSU. This 1.8V is the input to a Texas Instrument charge pump device on the motherboard that provides a 3V regulated output (TPS60303).

5.2.2.2 Use the CAA to Provide Power to the RPM

The regulated 1.8V supply from the CAA can be used to provide BelaSigna 300 with a 1.8V power source by bypassing the on-board LDO on the RPM. Typically the BelaSigna 300 supply is provided by the on-board regulator that uses the motherboard 3V power source as an input. To disable the on-board regulator, do the following:

1. Configure the PSU-EN header to short pins 2 to 3. This grounds the 3V-RPM signal, which is connected on the RPM to the voltage regulator’s enable pin.
2. Short jumper VBAT-EN, allowing the CAA-PSU signal to go directly to the RPM’s VBAT pin, bypassing the RPM’s on-board voltage regulator.

5.2.3 Power LED

A power indicator LED on the motherboard confirms when the main power switch (SW1) is active, and that the motherboard is powered correctly. This LED is active in all the power supply options.

5.2.4 Measuring Power Consumption

Measuring the power consumption of BelaSigna 300 on the Evaluation and Development Board can be done by one of the following methods:

- RPM powered by the motherboard using the CAA power supply
- RPM powered directly from an external 1.8V source

Each method is described in more detail below.

Measuring the Power Consumption when the RPM is Connected to the Motherboard

Begin by following the instructions described in Section 5.2.2.2, “Use the CAA to Provide Power to the RPM” on page 12 to power the RPM with 1.8V directly from the motherboard. Next, do the following:

1. Disable the on-board RPM voltage regulator (LDO) by shorting pins 2 to 3 on PSU-EN.
2. Place the ampere meter on header VBAT-EN; it measures the overall power consumption on the VBAT node.
3. Note that a few motherboard components might impact the measurement. Disable them as follows to isolate the power consumption measurement to only BelaSigna 300:
    a. Ensure you remove all communication cables (I^2C or UART), since the CAA could alter the measurement.
    b. If headphones or speakers are connected to any of the output connectors, their power consumption is included in the measurement; therefore remove them to isolate the power consumption of BelaSigna 300.
    c. The EEPROM on the RPM is also using current on the VBAT pin, but it is usually not being used, and should therefore not have an impact on the measurement, unless BelaSigna 300 frequently accesses the EEPROM. This can be the case when using the EEPROM to store compressed audio, or when logging data on a regular basis to the non-volatile memory.

Measuring the Power Consumption when the RPM is Not Connected to the Motherboard

Remove the RPM from the motherboard and ensure that resistors R2, R3 and R4 are connected (this is the default). Do the following:

1. Apply a 1.8V signal to either pin 2 or pin 4 of header X2.
2. Use either pin 5, 6 or 8 of the same connector for ground.
3. Place an ampere meter in series with the 1.8V supply and header X2 pin 2 or 4.

5.3 CLOCKING OPTIONS

All available clocking options are described under Section 4.3, “Clocking Options” on page 8. No other options are available on the motherboard, other than those described at the RPM level. The EXT_CLK pin of BelaSigna 300 is available on header X2 pin 7, as well as on the header EXT_CLK (pin 1) on the motherboard. Use this header to provide a different clock source to BelaSigna 300 from an external clock generator equipment. When using a different clock source from the on-board oscillator you must make hardware modifications. Refer to Section 4.3, “Clocking Options” on page 8 for more information on the components to remove.

5.4 INTERFACES

5.4.1 Audio Input Connection

The motherboard has a stereo 3.5mm jack connector (MIC), allowing a stereo signal to be fed to BelaSigna 300. The left and right channels of that signal are routed to the AI0 and AI2 inputs. The motherboard also has RCA connectors giving access to each of the four audio input channels: AI0, AI1, AI2 and AI4.

The motherboard also provides 3-pin headers for the four inputs. They are labeled AI0H, AI1H, AI2H and AI4H and all allow two-terminal or three-terminal microphones to be connected to the motherboard. Biasing circuitry is provided on the RPM for two-terminal microphones (for AI0 and AI2 only). Using three-terminal microphones on these two channels requires the removal of resistors R11 and R12 from the RPM board; see Section 4.6.1, “Audio Input Stage” on page 9 for more details.

Ground the unused analog inputs, if necessary. For example, if AI4 is not being used, short pins 2 to 3 on AI4H to tie this input to ground. This minimizes the potential of cross-talk in the input channels.

5.4.2 Audio Output Stage

The unfiltered direct digital output can be accessed on several headers located on the motherboard: pins 1 and 3 on headers RC-FILT-EN, LC-FILT-EN, RCVR-SEL, and pins 2 and 4 on HP-EN. A 3.5mm stereo jack (OUT) can also be configured to access the direct digital output by configuring the OUT-SEL and SHIELD-SEL headers (refer to Table 6).

TABLE 6: OUT 3.5MM STEREO JACK CONFIGURATION

Function	OUT-SEL	SHIELD-SEL
DAC Output	short pins 1 to 2 short pins 3 to 4 open pins 5 to 6 open pins 7 to 8	short pins 2 to 3
Direct Digital Output	open pins 1 to 2 open pins 3 to 4 short pins 5 to 6 short pins 7 to 8	short pins 1 to 2

Two filters are provided on the motherboard to attenuate out-of-band noise from the direct digital output. An example LC filter is provided that offers better noise shaping and power consumption optimization, and an RC filter is also provided for connecting to high impedance audio measurement equipment.

To listen to the direct digital output, either the direct digital output or the output of the LC filter can be routed to the 3.5mm stereo jack (OUT) by configuring the RCVR-SEL header. Shorting pins 1 to 2 and 3 to 4 accesses the direct digital output with no filtering. Shorting pins 5 to 6 and 7 to 8 provides access to the output of the LC filter.

The on-board filters are enabled by either the LC-FILT-EN or RC-FILT-EN header as shown in Table 7. Enable only one filter at a time.

TABLE 7: ON-BOARD LC/RC FILTER CONFIGURATION

Function	LC-FILT-EN or RC-FILT-EN
Disabled	open pins 1 to 2 open pins 3 to 4
Enabled	short pins 1 to 2 short pins 3 to 4

The direct digital output can be configured to operate in high power mode. In high power mode, two H-Bridge stages are arranged in parallel to effectively reduce the output impedance by half and achieve higher maximum output. To use the high power mode, configure the HP-EN header as described in Table 8, and adjust the RCVR-SEL jumpers to enable or bypass the LC filter. See Table 23 on page 20 for more information about configuring header RCVR-SEL.

To reduce the output level of the direct digital output, use the Rgain potentiometer.

TABLE 8: HIGH POWER MODE CONFIGURATION

Function	HP-EN
Disabled	open pins 1 to 2 open pins 3 to 4
Enabled	short pins 1 to 2 short pins 3 to 4

For prototyping stereo output applications, or where analog outputs are needed, the motherboard includes BelaSigna 250 which functions as an audio DAC. The motherboard also features an integrated headphone amplifier. To see the detailed schematics of this analog output stage, refer to Figure 5 on page 31.

The PCM interface from BelaSigna 300 is routed to BelaSigna 250 which provides an analog signal to the headphone amplifier. The stereo audio can be heard on the 3.5mm stereo jack. Additionally, the analog outputs from the headphone amplifier can be accessed on standard 0.1" headers: AO0-AMP+ and AO1-AMP+. A volume control (VOL) adjusts the signal level from the output of the audio amplifier. The output of the audio amplifier is designed to drive 16Ω or 32Ω headphones.

5.4.3 I²C Communication Interface

The I²C port provides full access to the chip for communicating with and debugging BelaSigna 300 devices. This port is supported by our development tools, which are part of the BelaSigna 300 Evaluation and Development Kit.

The I2C_SDA and I2C_SCL pins are available on the 6-pin mini DIN port labelled J1. They are also available on the 4-pin header, I2C on pins 2 and 3.

I²C interface communication with the motherboard requires an ON Semiconductor CAA. You can typically use an ON Semiconductor CAA for I²C communication between a PC and BelaSigna 300. The CAA always senses the power supply from the motherboard, so that it can apply appropriate level translation. The VBAT pin on BelaSigna 300 provides the Vsense signal used by the CAA.

Headers X5 and X6 are also available in case a pull-up resistor needs to be applied to the clock and data lines of the I²C interface. These headers are not connected by default.

5.4.4 General-Purpose I/O

BelaSigna 300 has five general purpose I/O pins, which have multiplexed functionalities. For example, GPIO0 and GPIO1 can be used as standard general purpose I/O pins, but also as the general-purpose RS232 UART (see Section 5.4.5, “General-Purpose RS232 UART” on page 16 for details). Also, GPIO1 to GPIO4 can be multiplexed between standard GPIO function and low-speed A/D functionality.

All GPIO pins are routed to the motherboard which uses them all in GPIO mode. Specifically, GPIO0 and GPIO1 are connected to toggle switches that can be used control algorithm parameters. These two toggle switches connect their associated GPIO to ground when they are in the low position (when you are looking at the motherboard with the RPM on its left). They are floating when the button is in the high position, and tied to the positive power supply of BelaSigna 300 because of its internal pull-ups.

Momentary push buttons are connected to GPIO2 and GPIO3; they are also designed to provide some control of the algorithms running on BelaSigna 300. When pressed, the momentary switches tie their associated GPIOs to ground. When released, the GPIOs are floating again, and pulled up internally by BelaSigna 300.

An LED (D2) on the motherboard is connected to GPIO4. When a DSP application configures GPIO4 as an output, it drives the LED on and off, depending on the GPIO state. This is a convenient method of showing the behavior of any algorithm function running on BelaSigna 300. To use this functionality, short the LED-EN header.

5.4.5 General-Purpose RS232 UART

A general-purpose UART connection is available on the 10-pin header called RS232. To use this header, short pins 1 to 2 and 3 to 4. This header is designed for compatibility with the ribbon cables provided by ON Semiconductor with hardware communication interfaces. This interface does not contain any on-board level translation; consequently, the signal levels are those defined by BelaSigna 300, typically 1.8V. Note that this interface cannot be used to provide power to the motherboard.

5.4.6 PCM/I²S Interface

The PCM/I²S interface of BelaSigna 300 is provided on a single header called PCM. This interface can be useful for high-speed bi-directional data transfers. On the motherboard, the PCM interface sends the audio data to a stereo DAC/amplifier allowing a stereo output signal to be heard on the output jack connector. See Section 5.4.2, “Audio Output Stage” on page 14 for more information on using the stereo DAC.

A Configuration Options

A.1 Available Signals on the Motherboard

Table 9: Header AI0H—Connect a Microphone on the First Channel

Designator	Signal Name	Description
1	VREG	1.0V regulated supply for microphone biasing
2	AI0	Connects to AI0 analog input through 100nF capacitor
3	GNDA	Microphone ground

Table 10: Header AI1H—Connect a Microphone on the Second Channel

Designator	Signal Name	Description
1	VREG	1.0V regulated supply for microphone biasing
2	AI1	Connects to AI1 analog input through 100nF capacitor
3	GNDA	Microphone ground

Table 11: Header AI2H—Connect a Microphone on the Third Channel

Designator	Signal Name	Description
1	VREG	1.0V regulated supply for microphone biasing
2	AI2	Connects to AI2 analog input through 100nF capacitor
3	GNDA	Microphone ground

Table 12: Header AI4H—Connect a Microphone on the Fourth Channel

Designator	Signal Name	Description
1	VREG	1.0V regulated supply for microphone biasing
2	AI4	Connects to AI4 analog input through 100nF capacitor
3	GNDA	Microphone ground

Table 13: Header RCVRH—Provides Access to the Filtered Direct Digital Output

Position	Signal Name	Description
left	RCVR-	Direct digital output - (after an RC filter)
right	RCVR+	Direct digital output + (after an RC filter)

TABLE 14: HEADER EXTCLK—PROVIDES ACCESS TO THE EXTERNAL CLOCK PIN

Position	Signal Name	Description
top (+)	EXTCLK	External clock provided to BelaSigna 300
bottom	GND	Ground

TABLE 15: HEADER I2C—PROVIDES ACCESS TO THE I²C INTERFACE

Designator	Signal Name	Description
1	-	-
2	SCL	I²C clock signal
3	SDA	I²C serial data
4	GND	Ground

TABLE 16: HEADER X5—ALLOWS CONNECTION TO A PULL-UP RESISTOR ON THE SCL PIN

Position	Signal Name	Description
1	1V8	I²C interface power supply (VDDC)
2	SCL	I²C clock signal

TABLE 17: HEADER X6—ALLOWS CONNECTION TO A PULL-UP RESISTOR ON THE SDA PIN

Position	Signal Name	Description
1	1V8	I²C interface power supply (VDDC)
2	SDA	I²C serial data

TABLE 18: HEADER PCM—PROVIDES ACCESS TO THE PCM INTERFACE

Designator	Signal Name	Description
1	PCM-FR	PCM frame signal
2	PCM-SI	PCM serial input
3	PCM-SO	PCM serial output
4	PCM-CLK	PCM clock signal
5	GND	Ground

TABLE 19: HEADER GPIO—PROVIDES ACCESS TO GPIO SIGNALS

Designator	Signal Name	Description
1	GND	Digital ground
2	GPIO0	GPIO0
3	GND	Ground
4	GPIO1	GPIO1/LSAD1
5	GND	Ground
6	GPIO2	GPIO2/LSAD2
7	GND	Ground
8	GPIO3	GPIO3/LSAD3
9	GND	Ground
10	GPIO4	GPIO4/LSAD4

A.2 CONFIGURATION HEADERS

TABLE 20: HEADER VBAT-EN—DIRECT 1.8V SUPPLY TO THE RPM

Designator	Signal Name	Description
1	CAA-PSU	Direct 1.8V supplied provided by the CAA
2	1V8	RPM VBAT supply

By default, header VBAT-EN is *not* connected. When connected, this function powers the RPM directly with 1.8V provided by the CAA, and bypasses the on-board regulator on the RPM. When connecting this header, *ensure that header PSU-EN shorts pins 2 to 3*.

TABLE 21: HEADER PSU-EN—PROVIDE 3V TO THE ON-BOARD LD0 ON THE RPM

Designator	Signal Name	Description
1	PSU	Motherboard's 3V
2	3V-RPM	Input to RPM LD0
3	GND	Ground

By default, header PSU-EN has pins 1 and 2 shorted. To provide the 3V power to the RPM's voltage regulator, you must also *disconnect the header VBAT-EN*.

To disable the RPM's voltage regulator, short pins 2 and 3 on header PSU-EN, and ensure that you *connect the header VBAT-EN* so that the RPM still gets 1.8V of power.

TABLE 22: HEADER PSU-SEL—SELECT THE POWER SOURCE TO THE MOTHERBOARD

Designator	Signal Name	Description
1	battery	Battery power supply
2	psu-sel	Motherboard 3V power supply
3	3VCP	Charge pump 3V supply
4	psu-sel	Motherboard 3V power supply

By default, header PSU_SEL has pins 1 and 2 shorted, providing power to the motherboard through the AAA batteries. To select the other power option, short pins 3 to 4 instead.

TABLE 23: HEADER RCVR-SEL—SELECT THE FILTERED OR UNFILTERED OUTPUT

Designator	Signal Name	Description
1	RCVR-	Unfiltered direct drive output -
2	OUT-	Selected output -
3	RCVR+	Unfiltered direct drive output +
4	OUT+	Selected output +
5	LC-OUT-	LC-filtered direct drive output -
6	OUT-	Selected output -
7	LC-OUT+	LC-filtered direct drive output +
8	OUT+	Selected output +

By default, header RCVR_SEL has pins 5 to 6 and 7 to 8 shorted, providing the output of the on-board LC filter to the output connectors. To select the unfiltered direct drive output, short pins 1 to 2 and 3 to 4 instead.

TABLE 24: HEADER OUT-SEL—SELECT THE DIRECT DRIVE OUTPUT OR THE EXTERNALLY CONVERTED AND AMPLIFIED OUTPUT

Designator	Signal Name	Description
1	A0-AMP	Externally converted and amplified left output
2	tip	3.5mm tip signal (left)
3	A1-AMP	Externally converted and amplified right output
4	ring	3.5mm ring signal (right)
5	OUT+	Direct drive output +
6	tip	3.5mm tip signal (left)
7	OUT+	Direct drive output +
8	ring	3.5mm ring signal (right)

By default, header OUT_SEL has pins 1 to 2 and 3 to 4 shorted, providing externally converted and amplified output. To have the direct drive output go to the output jack connectors, short pins 5 to 6 and 7 to 8 instead.

TABLE 25: HEADER RC-FILT-EN—ENABLE THE RC FILTER NETWORK

Designator	Signal Name	Description
1-2	RCVR-	Direct drive output -
3-4	RCVR+	Direct drive output +

By default, header RC-FILT-EN is *not connected*. To route the direct drive output through the RC filter network, short pins 1 to 2 and 3 to 4.

TABLE 26: HEADER LC-FILT-EN—ENABLE THE LC FILTER NETWORK

Designator	Signal Name	Description
1-2	RCVR-	Direct drive output -
3-4	RCVR+	Direct drive output +

By default, header LC-FILT-EN is *connected*. To disconnect the direct drive output from the LC filter network, open pins 1 to 2 and 3 to 4.

TABLE 27: HEADER HP-EN—ENABLE THE HIGH-POWER OUTPUT

Designator	Signal Name	Description
1	RCVR_HP-	High power output -
2	RCVR-	Direct drive output -
3	RCVR_HP+	High power output +
4	RCVR+	Direct drive output +

By default, header HP-EN is *not connected*. To enable high-power output on the direct output driver, short pins 1 to 2 and 3 to 4.

TABLE 28: HEADER LED-EN—ENABLE LED D2 ON GPIO4

Designator	Signal Name	Description
left	GPIO4	GPIO4/LSAD4
right	-	-

By default, header LED-EN is *not* connected. When connected, this function connects GPIO4 to the D2 LED, enabling this GPIO (when configured as an output) to drive the LED.

TABLE 29: HEADER MIC-PSU

Designator	Signal Name	Description
1	VREG	Short pins 1 to 2
2	1.8V	Short pins 3 to 4
3	3V	Short pins 5 to 6

TABLE 30: HEADER BS250-EN

Designator	Signal Name	Description
top	1V8	1.8V power supply source
bottom	BS250-VBAT	VBAT pin to BelaSigna 250

CAUTION: By default, header BS250-EN is connected to use the on-board DA converter. Do not disconnect.

TABLE 31: HEADER RXTX-SEL

Designator	Signal Name	Description
	UART Mode	Short pins 1 to 2 Short pins 3 to 4
	Debug Mode	Short pins 5 to 6 Short pins 7 to 8

A.3 SWITCHES

The motherboard has five switches. The main switch (SW1) controls the power supply of the board, whereas the others are linked to GPIOs. Switches GPIO0 and GPIO1 are toggle switches, and GPIO2 and GPIO3 are momentary switches (buttons).

TABLE 32: SWITCH SW1—MAIN POWER SWITCH

Designator	Signal Name	Description
OFF	-	In this position, no power is applied to the motherboard.
ON	psu-sel	In this position, the power source selected by pin psu-sel is provided as general power supply to the motherboard.

TABLE 33: SWITCH GPIO0—TOGGLE SWITCH FOR GPIO0

Designator	Signal Name	Description
High	-	In this position, GPIO0 is left floating; connection to VDD is provided by internal pull-ups.
Low	GND	In this position, GPIO0 is tied to ground.

TABLE 34: SWITCH GPIO1—TOGGLE SWITCH FOR GPIO1

Designator	Signal Name	Description
High	-	In this position, GPIO1 is left floating; connection to VDD is provided by internal pull-ups.
Low	GND	In this position, GPIO1 is tied to ground.

TABLE 35: SWITCH GPIO2—MOMENTARY SWITCH FOR GPIO2

Designator	Signal Name	Description
Unpressed	-	When not pressed, GPIO2 is left floating; connection to VDD is provided by internal pull-ups.
Pressed	GND	While the button is pressed, GPIO2 is tied to ground. Releasing the button puts the signal back to VDD.

TABLE 36: SWITCH GPIO3—MOMENTARY SWITCH FOR GPIO3

Designator	Signal Name	Description
Unpressed	-	When not pressed, GPIO3 is left floating; connection to VDD is provided by internal pull-ups.
Pressed	GND	While the button is pressed, GPIO3 is tied to ground. Releasing the button puts the signal back to VDD.

A.4 POTENTIOMETERS

The motherboard has a potentiometer that controls the volume on the headphone jack. Only the analog output level from the audio amplifier is connected by this potentiometer. Another potentiometer adjusts the level of the direct drive output.

TABLE 37: POTENTIOMETER VOL—OUTPUT VOLUME CONTROL

Description
Adjusts the volume of the headphone output from BelaSigna 250 and the headphone amplifier. The volume ranges from Mute to Full scale signal.

TABLE 38: POTENTIOMETER RGAIN—DIRECT DRIVE OUTPUT CONTROL

Description
Controls the volume of the RCVR direct drive output level from BelaSigna 300.

A.5 CONNECTORS

TABLE 39: CONNECTOR MIC—ANALOG INPUT 3.5MM JACK

Designator	Signal Name	Description
1	GNDA	Analog ground
2	ai2	Analog input
3	ai0	Analog input

TABLE 40: CONNECTOR OUT—ANALOG OUTPUT 3.5MM JACK

Designator	Signal Name	Description
1	SHIELD-SEL	Signal selected by the SHIELD-SEL header (Ground or OUT-)
2	OUT-SEL2	Connects to OUT-SEL pins 4 and 8 (AO0-AMP or OUT+)
3	OUT-SEL1	Connects to OUT-SEL pins 2 and 6 (AO1-AMP or OUT+)

TABLE 41: CONNECTOR AI0—RCA CONNECTOR TO FIRST CHANNEL

Designator	Signal Name	Description
inner	ai0	Analog input to first channel
outer	GNDA	Analog ground

TABLE 42: CONNECTOR AI1—RCA CONNECTOR TO SECOND CHANNEL

Designator	Signal Name	Description
1	ai1	Analog input to second channel
2	GNDA	Analog ground

TABLE 43: CONNECTOR AI2—RCA CONNECTOR TO THIRD CHANNEL

Designator	Signal Name	Description
1	ai2	Analog input to third channel
2	GNDA	Analog ground

TABLE 44: CONNECTOR AI4—RCA CONNECTOR TO FOURTH CHANNEL

Designator	Signal Name	Description
1	ai4	Analog input to fourth channel
2	GNDA	Analog ground

TABLE 45: CONNECTOR J1—I²C MINI DIN CONNECTOR

Description		
1	CAA-PSU	Power supply provided by the CAA (1.8V)
2	GND	Ground
3	SCL	I²C clock signal
4	SDA	I²C serial data line
5	1V8	1.8V power supply
6	-	Not connected

TABLE 46: CONNECTOR CAA—10-PIN RS232 COMMUNICATION FOR UART

Designator	Signal Name	Description
1	1V8	1.8V power supply
2	GND	Ground
3	uart-tx	UART transmit pin
4	GND	Ground
5	uart-rx	UART receive pin
6	GND	Ground
7	-	Not connected
8	GND	Ground
9	-	Not connected
10	GND	Ground

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B

Schematics

This appendix contains schematics for the motherboard and the RPM.

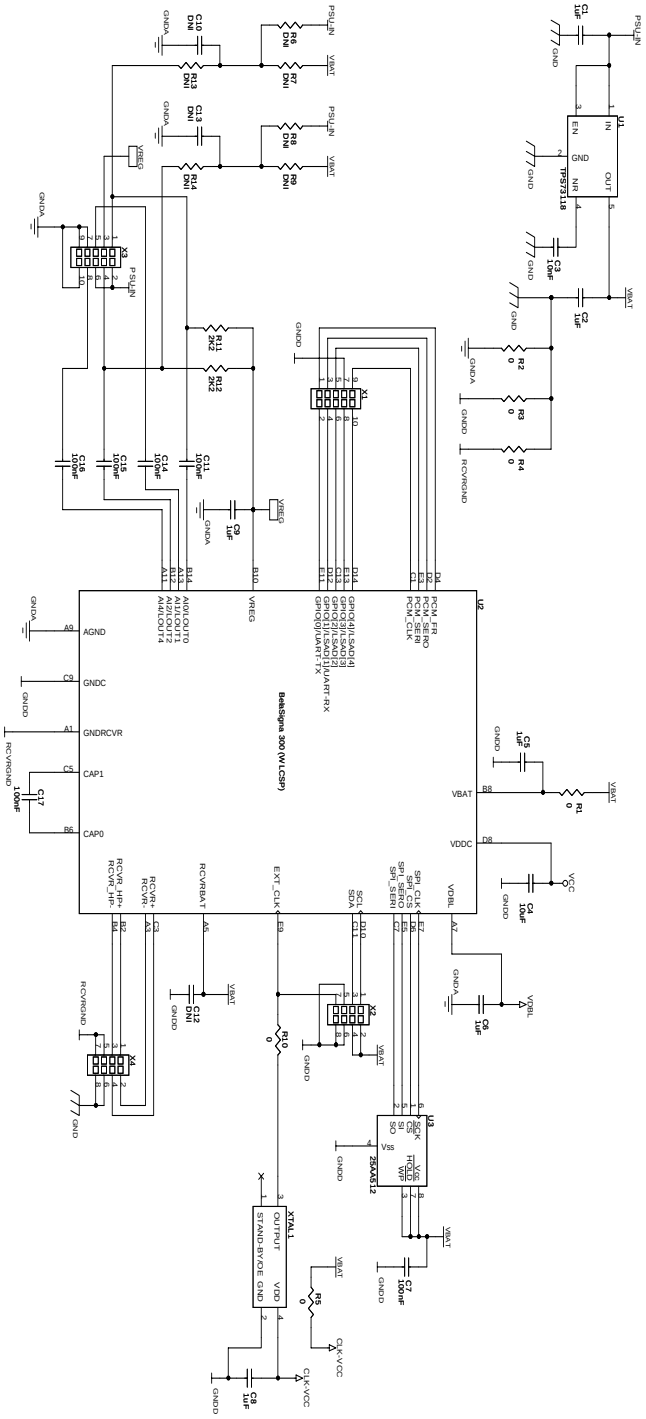


Figure 2: RPM Schematic

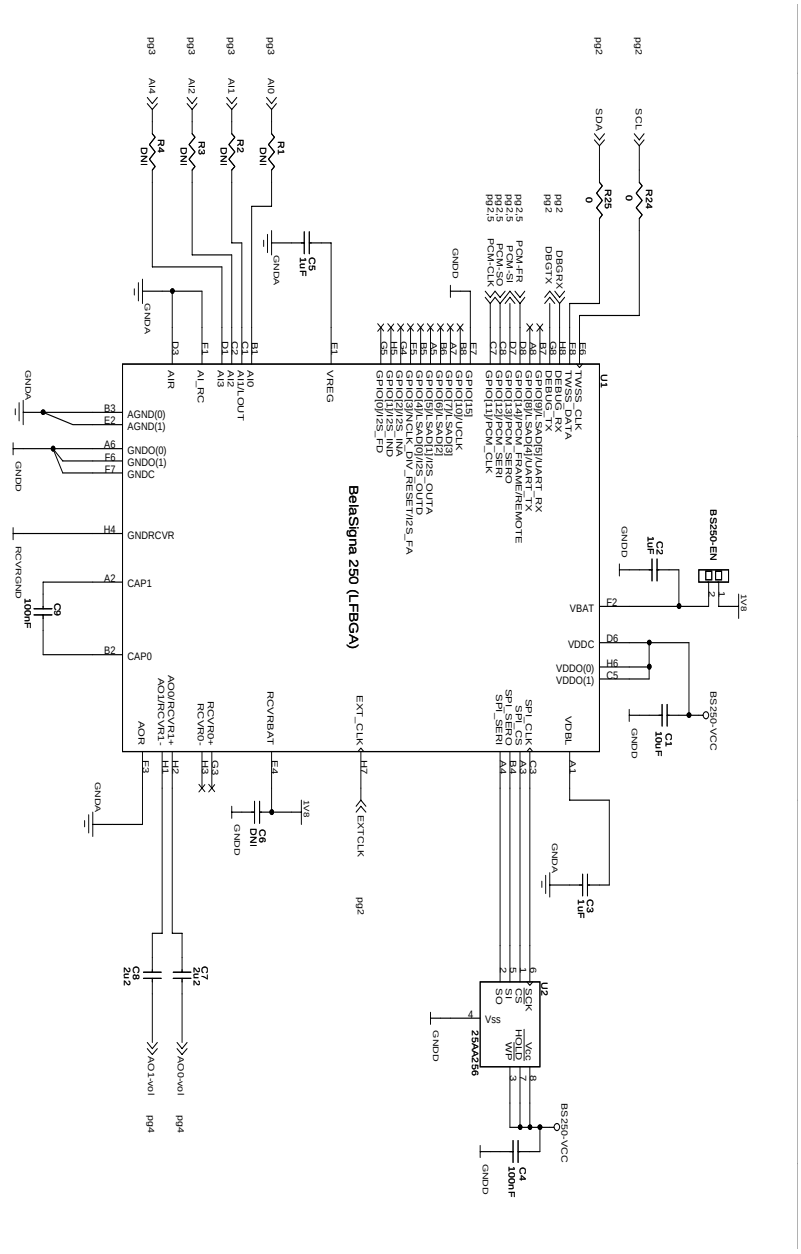


Figure 3: Motherboard Schematic—BelaSigna 250

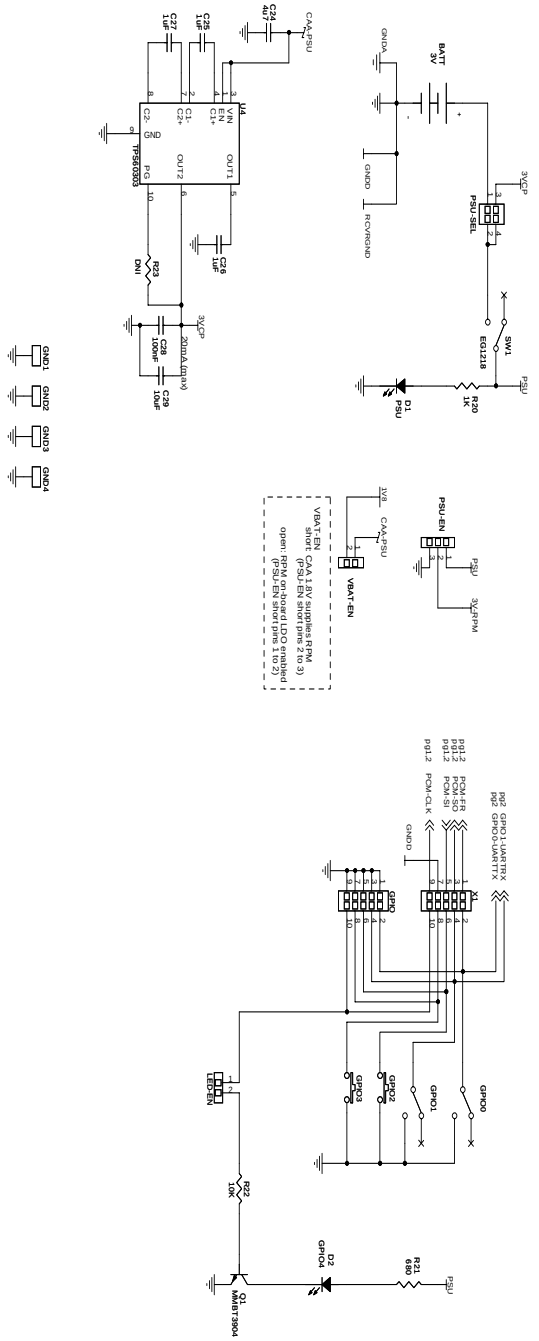


Figure 4: Motherboard—Power Supply and GPIOs

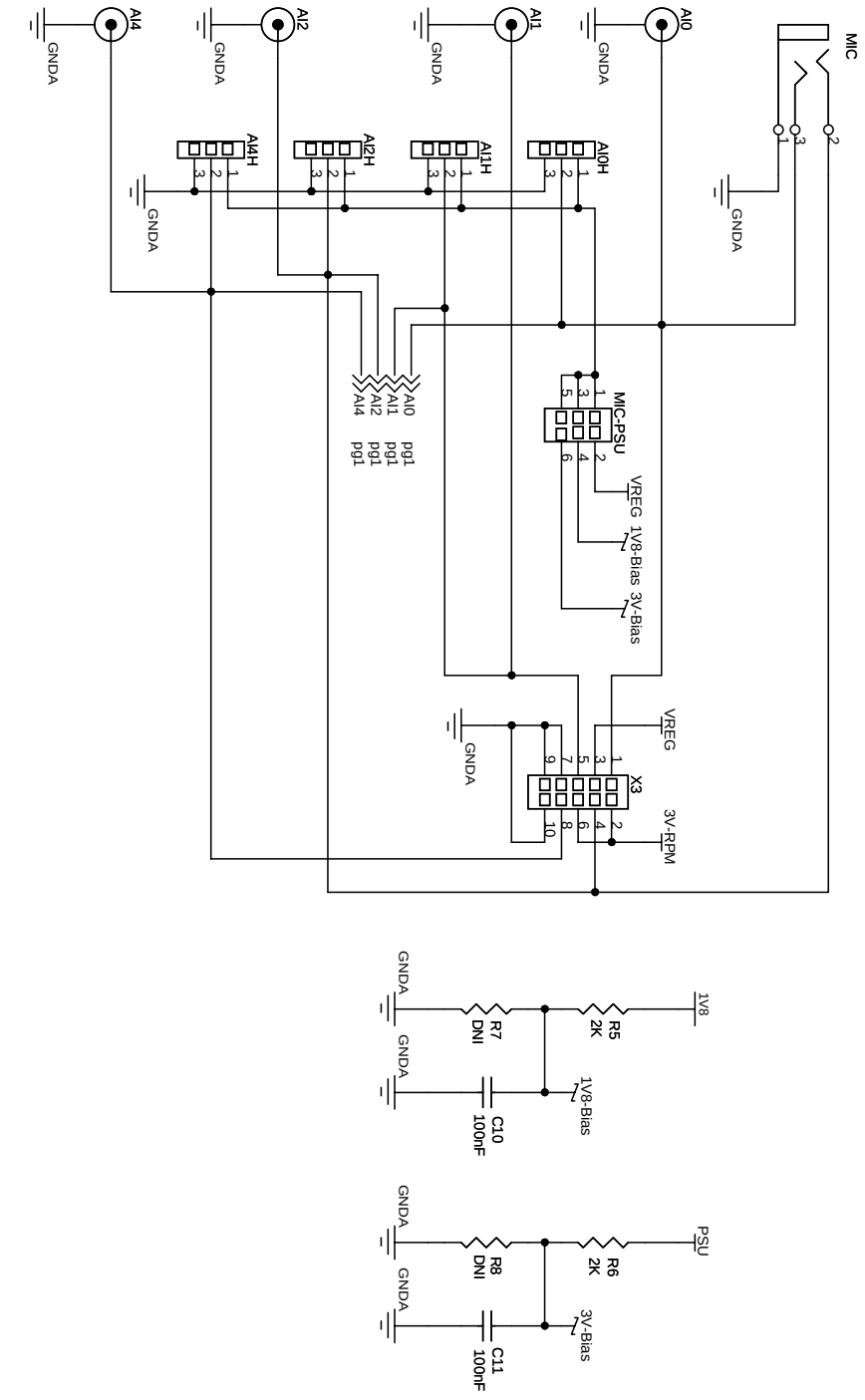


Figure 5: Motherboard—Input Stage

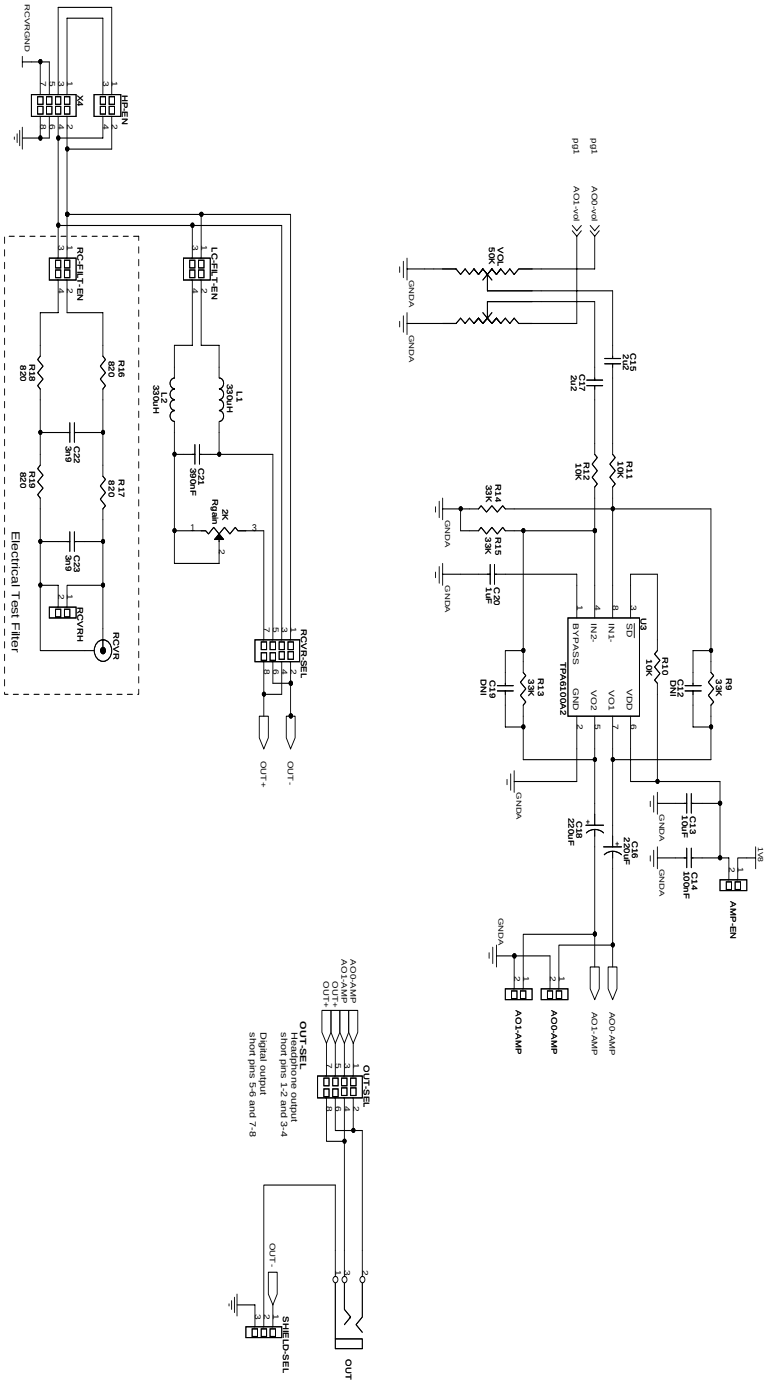


Figure 6: Motherboard—Output Stage

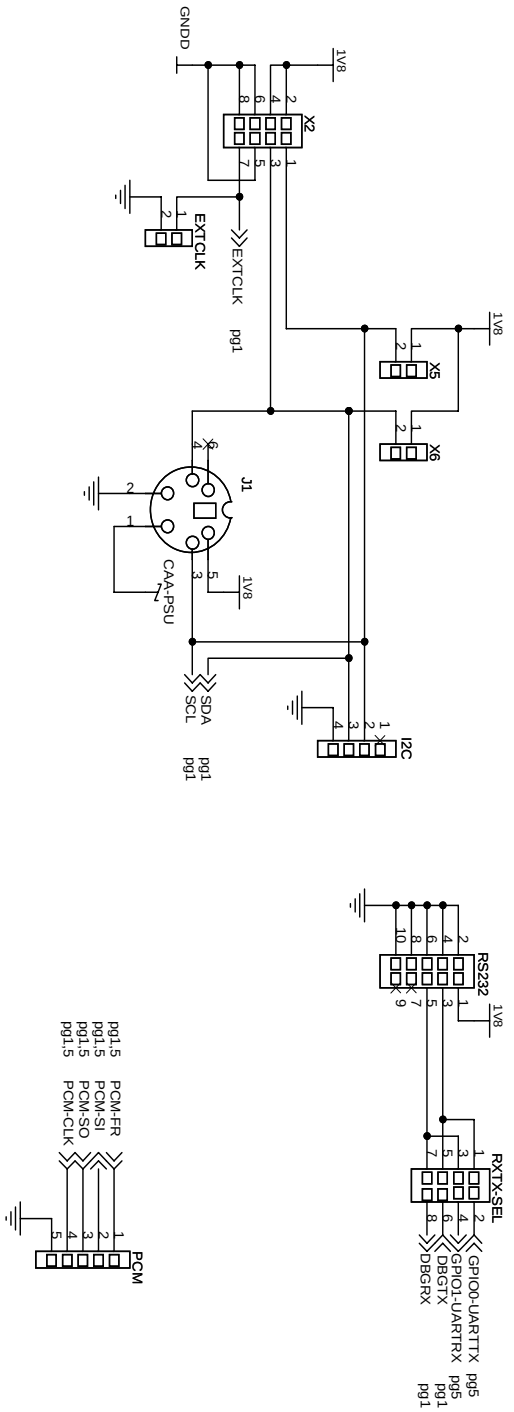


Figure 7: Motherboard—Communications Interface

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C Layout

C.1 RPM LAYOUT

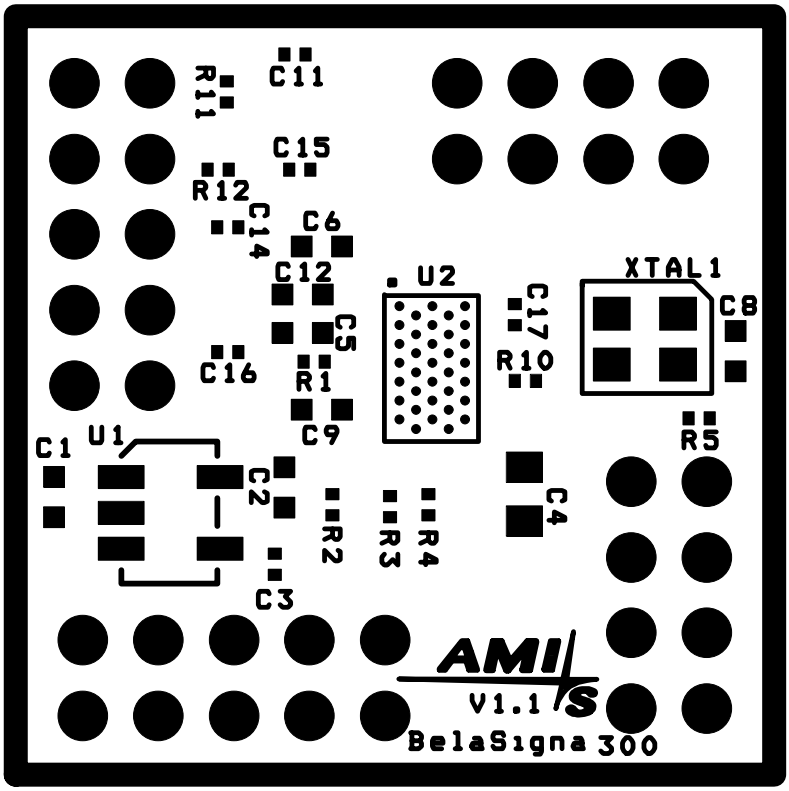


Figure 8: RPM Top Assembly Plan and Silkscreen (5:1 scale)

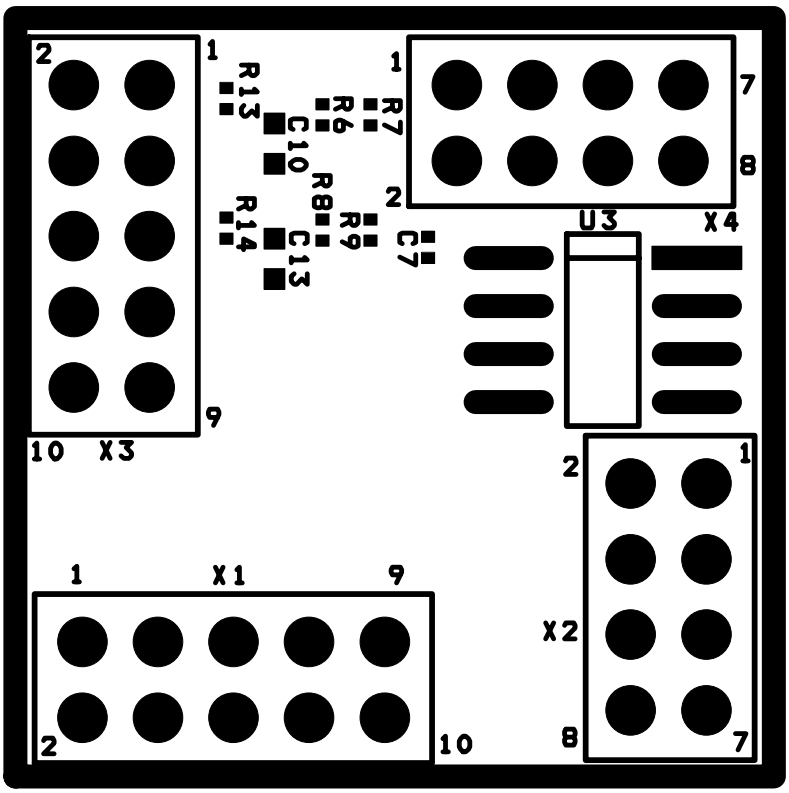


Figure 9: RPM Bottom Assembly Plan and Silkscreen (5:1 scale)

C.2 MOTHERBOARD LAYOUT

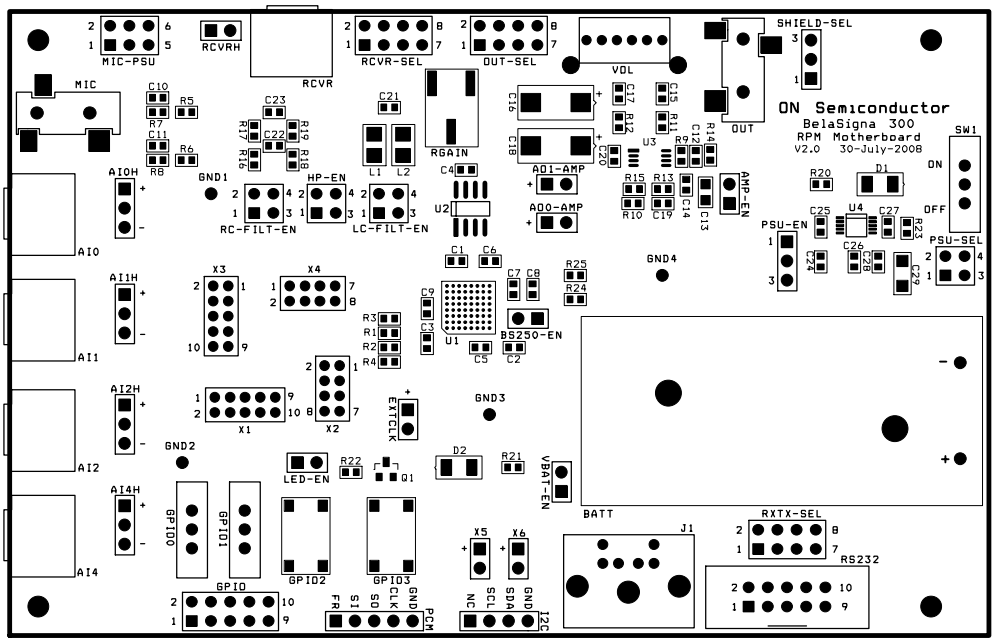


Figure 10: Motherboard Assembly Plan and Silkscreen

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